

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	Rising Waters
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Registration	Allows new Disaster Officers to securely create an official account.	Auth Service	Done	1 Mark
2	Officer Login & MFA	Validates registered credentials securely with multi-factor tokens.	Auth Service	Done	1 Mark
3	Live Telemetry Fetch	Ingests real-time water level data streams from regional IoT gauges.	Data Ingestion	Done	1 Mark
4	ML Flood Prediction	Executes Python predictive logic to estimate risk within 4-6 hours.	Analytics Engine	Done	1 Mark
5	Dynamic Risk Dashboard	Renders live localized risk heatmaps and historical charts on web interface.	Presentation Layer	Done	1 Mark
6	Twilio SMS Warning Relay	Dispatches automated flash warning mobile text messages to citizens.	Notification Service	Done	1 Mark
7	Multi-Language Voice Alerts	Broadcasts automated voice calls in local regional languages during emergency failures.	Notification Service	In Progress	1 Mark
8	Offline Maps Cache	Syncs cached offline area maps for ground-level field rescue teams.	Presentation Layer	Pending	1 Mark

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	6
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	2

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Interactive Risk Dashboard, Historical Precipitation Graphs
2	Backend / Logic	2	ML Risk Prediction Logic Engine, Telemetry Ingestion Handler
3	Database / Storage	2	MongoDB Live Sensor Logging, PostgreSQL Historical Records
4	API / Integration	2	OpenWeatherMap API Fetcher, Twilio SMS Notification Endpoint
5	Security / Authentication	2	Multi-Factor Officer Login, Secure Session JWT Generation